

CPP - Pre Req

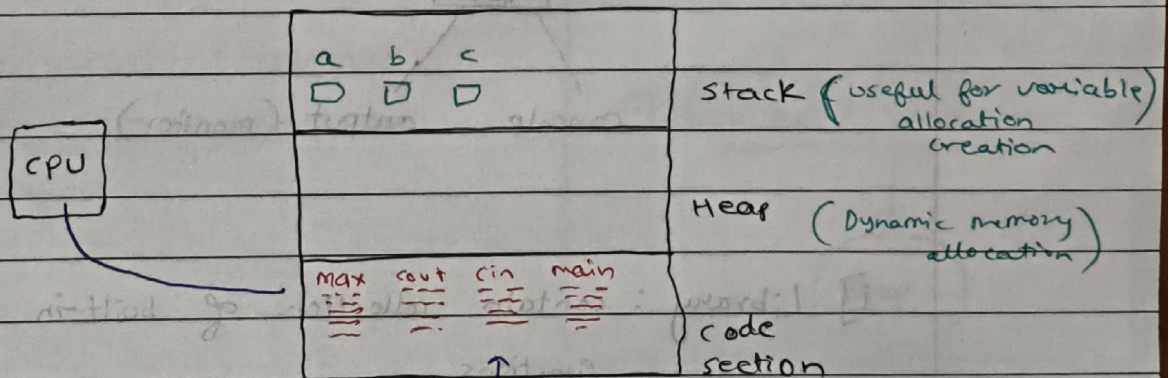
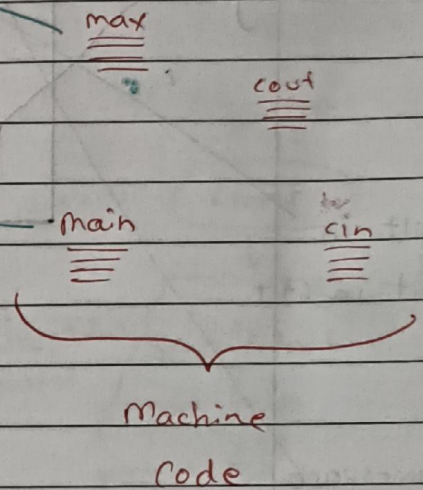
Steps for Program development and execution:-

first.cpp

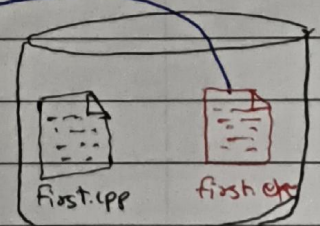
```
#include <iostream>
int max (int x, int y)
{
    if (x > y)
        return x;
    else
        return y;
}
```

```
int main()
{
    int a=10, b=5, c;
    c = max(a,b);
    std::cout << c;
    return 0;
}
```

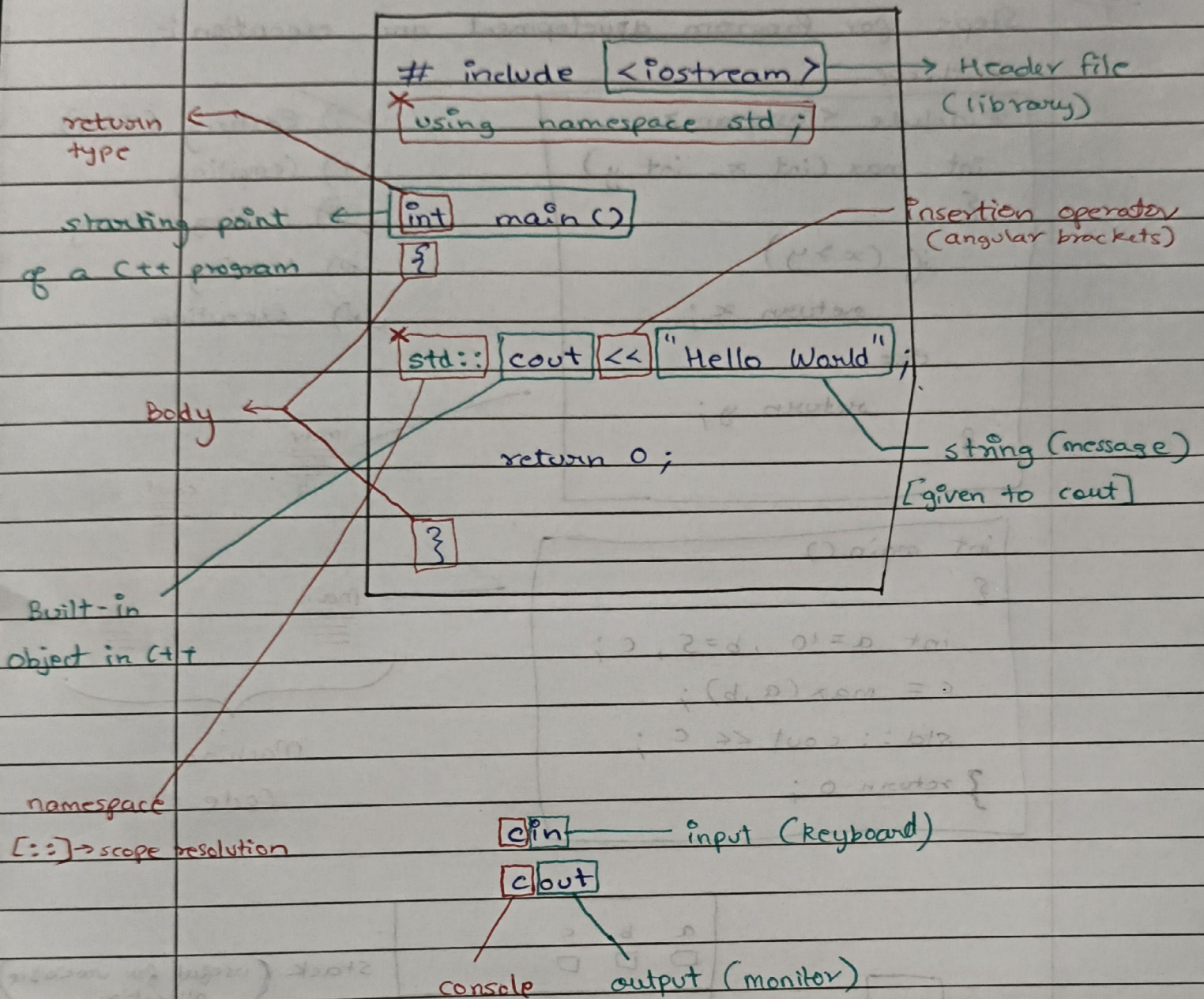
- ✓ i) Editing (IDE)
- ✓ ii) Compiling
- ✓ iii) Linking Library
- ✓ iv) Loading
- ✓ v) Execution



loading
done by os

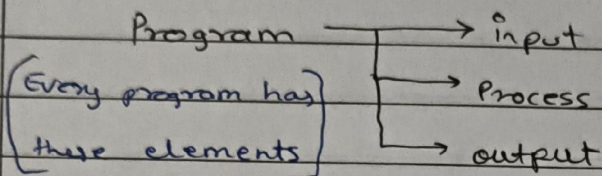


Skeleton of C++ Program



i] Library : contain collection of built-in objects and functions.

Writing first Program :-



Adding 2 numbers

① Algorithm (Pseudocode)

Begin

Print "Enter 2 nos"

Read a, b

$c \leftarrow a + b$

Print "Addition is", c

end

CODE

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c;
    cout << "Enter 2 numbers";
    cin >> a >> b;
    c = a + b;
    cout << "Addition is" << c;

    return 0;
}
```

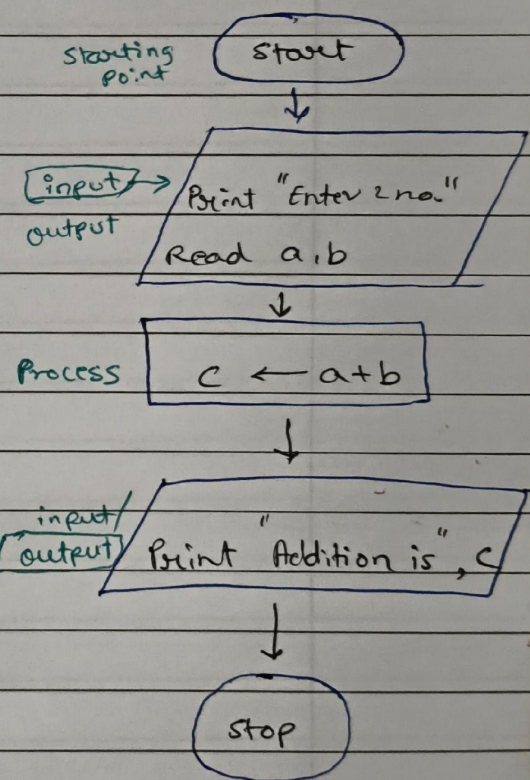


→ You cannot directly use variables without declaring

→ "=" here is not equal to. Read it as "assigned".

main() is mandatory as entry point of program

flowchart



getline (cin, str)

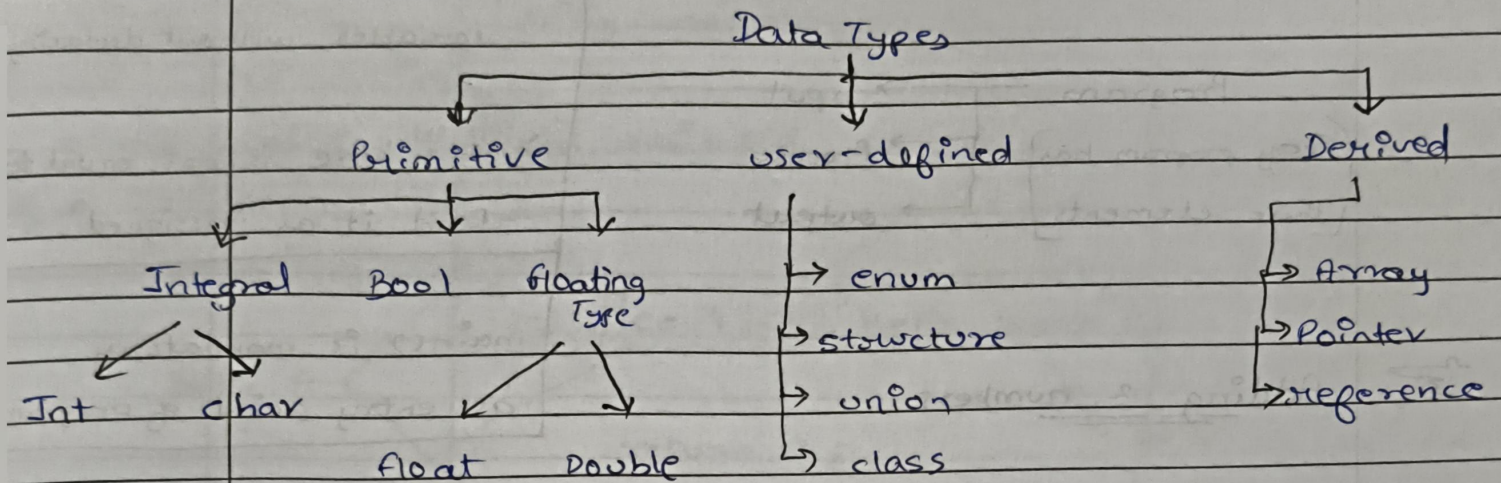
reads whole line of text including spaces

variable

where to store

where to read from

3 Data types & Variables → name given to data



`int rollNo = 10 ;`

`char group = 'A' ;`

`float price = 12.75f`

→ If not writing 'f', by default it will be considered double

3 Operators and Expressions :-

◆ Arithmetic Operators : + - * / %

asterisk

modulus

Divide gives Quotient

modulus gives remainder

cannot perform modulus on float. only on int & char

! every char = ASCII code, so operations are performed on their integral values.

→ Typecasting

```
int a=13, b=5;  
float c;
```

Typecasting ←

```
c = (float) a/b;
```

→ Expression Precedence :-

	Operator	Assumed	Precedence
Parenthesis →	()	3	Highest
	!, *, %	2	
	+, -	1	Lowest

#include <cmath> → C++ style

#include <math.h> → C style

→ Compound Assignments

+=

--

*=

/=

%=

```
int a=10, b=5, c=5;
```

```
int sum = 5;
```

```
sum = sum + a
```

OR

```
sum += a;
```

faster
execution